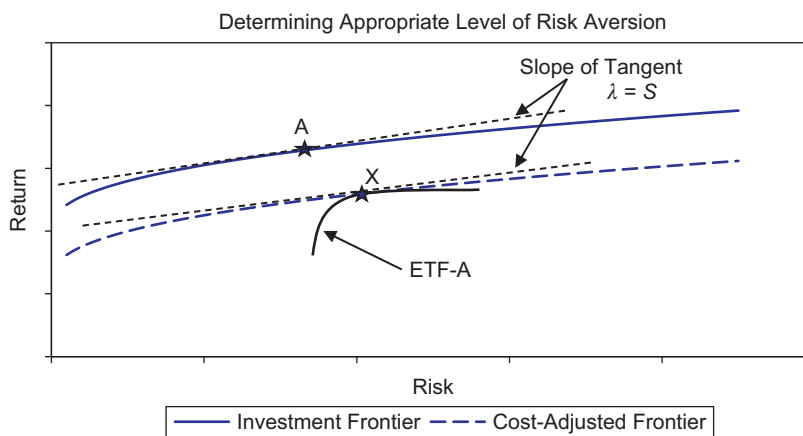


Investors will always seek out the highest efficient investment portfolio. Engle and Ferstenberg (2006, 2007) provide an alternative discussion of the cost-adjusted frontier. In their article, the authors present a framework to incorporate transaction costs directly into the investment process in order to determine a more efficient ex-post portfolio.

DETERMINING THE APPROPRIATE LEVEL OF RISK AVERSION

Suppose a manager constructs a portfolio by maximizing investor utility and then submits the list to the trader for execution. In most situations the trader does not have sufficient time or tools to perform a detailed cost analysis to determine the appropriate cost-adjusted frontier and corresponding execution strategy. However, the trader usually does have sufficient time to perform a single trade cost optimization as defined in Equation 10.4. But how should the trader specify the level of risk aversion to ensure the trading decision is consistent with the investment decision?

A joint examination of the efficient investment frontier and the cost-adjusted frontier provides some insight into our question (Figure 10.7). In the figure A represents the selected optimal portfolio and X represents the single best execution strategy. The question now shifts to finding this strategy. If we assume that all investors are indeed rational investors then the tangent to the efficient investment frontier at the optimal portfolio A is equal to the Sharpe ratio S of the portfolio (Sharpe, 1966),



■ **Figure 10.7** Determining the Appropriate Level of Risk Aversion. Source: *Journal of Trading* (2007).